S.Y.B.Sc.(Data Science) Sem- III

Subject: Data Structure – I

Unit2 : Array as a Data Structure

By,

Mrs. Reshma Masurekar

Dr. D. Y. Patil ACS College,

Pimpri, Pune 18.

Sorting

- Sorting is the process of arranging elements of a list in a specific order, based on a defined criteria.

Internal Sorting :

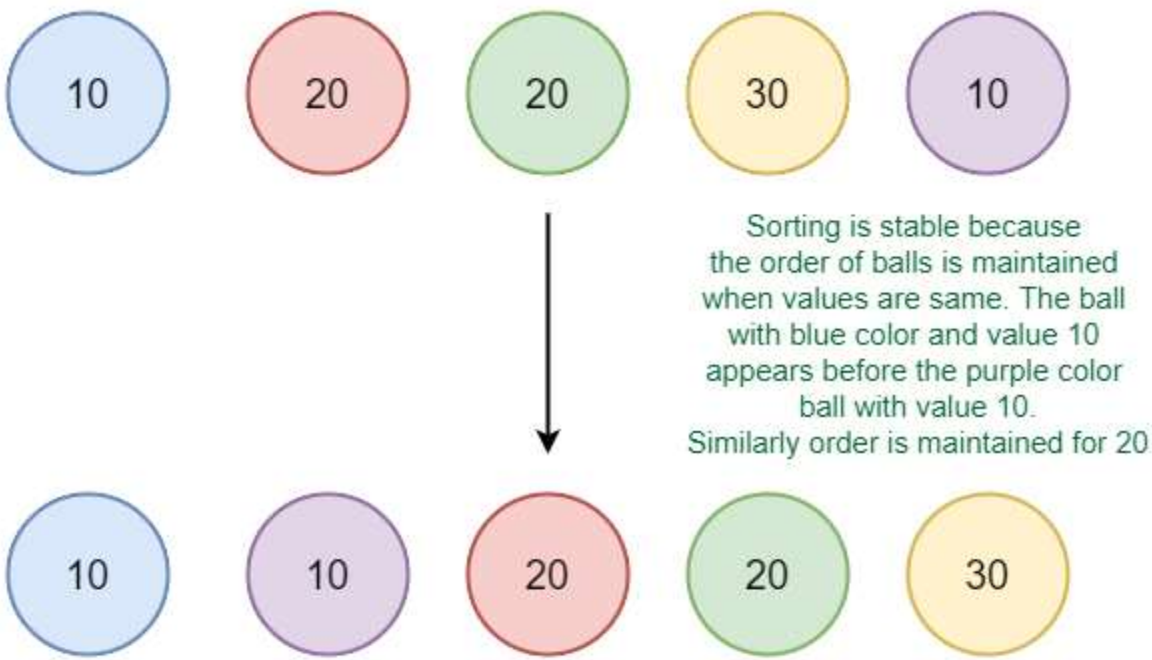
- When all data is placed in the **main memory** or **internal memory** then sorting is called internal sorting.
- In internal sorting, the problem cannot take input beyond its size.
- Example: bubble sort, selection sort, quick sort, shell sort, insertion sort, heap sort

External Sorting :

- When all data that needs to be sorted **cannot be placed in memory at a time**, the sorting is called external sorting.
- External Sorting is used for the massive amount of data.
- Merge Sort and its variations are typically used for external sorting.
- Some external storage like hard disks and CDs are used for external sorting.
- Example: Merge sort, radix sort, etc.

Stable sorting:

- When two same data appear in the **same order** in sorted data without changing their position is called stable sort.
- Example: *merge sort, insertion sort, bubble sort.*



In-Place Sorting:

- In-place sorting refers to sorting algorithms that rearrange the elements of an array directly within its original memory space, without needing extra space proportional to the input size. Essentially, they modify the existing array to produce the sorted output, rat
- An in-place sorting algorithm uses **constant space** for producing the output (modifies the given array only. Examples: Selection Sort, Bubble Sort, Insertion Sort and Heap Sort. her than creating a copy.

Types of Sorting Techniques

The following two types of sorting algorithms can be broadly classified:

- **Comparison-based:** We compare the elements in a comparison-based sorting algorithm)
- **Non-comparison-based:** We do not compare the elements in a non-comparison-based sorting algorithm)